

SUPPLEMENT 2

DERIVATION OF SIMPLE REGRESSION FORMULAS FOR SLOPE AND Y INTERCEPT

Derivation of the formulas for the slope and y intercept involve using calculus partial derivative techniques to minimize the sum of squares of error (SSE):

$$SSE = \sum (y - \hat{y})^2 \text{ but } \hat{y} = b_0 + b_1x$$

$$SSE = \sum [y - (b_0 + b_1x)]^2$$

$$SSE = \sum (y - b_0 - b_1x)^2$$

Taking the partial derivative of SSE with respect to b_0 and setting it equal to 0 (to minimize):

$$\frac{\partial SSE}{\partial b_0} = 2 \sum (y - b_0 - b_1x)^1 (-1) = 0$$

$$\sum (y - b_0 - b_1x) = 0$$

$$\sum y - \sum b_0 - b_1 \sum x = 0$$

$$\sum y - b_0 n - b_1 \sum x = 0$$

$$b_0 n = \sum y - b_1 \sum x$$

$$b_0 = \frac{\sum y}{n} - b_1 \frac{\sum x}{n}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Taking the partial derivative of SSE with respect to b_1 and setting it equal to 0 (to minimize):

$$\frac{\partial SSE}{\partial b_1} = 2 \sum (y - b_0 - b_1 x)^1 (-x) = 0$$

$$\sum (xy - b_0 x - b_1 x^2) = 0$$

$$\sum xy - b_0 \sum x - b_1 \sum x^2 = 0$$

$$\text{but } b_0 = \frac{\sum y}{n} - b_1 \frac{\sum x}{n}$$

$$\sum xy - \left(\frac{\sum y}{n} - b_1 \frac{\sum x}{n} \right) \sum x - b_1 \sum x^2 = 0$$

$$\sum xy - \frac{\sum x \sum y}{n} + b_1 \frac{(\sum x)^2}{n} - b_1 \sum x^2 = 0$$

$$\frac{\sum x \sum y}{n} - \sum xy + b_1 \sum x^2 - b_1 \frac{(\sum x)^2}{n} = 0$$

$$b_1 \sum x^2 - b_1 \frac{(\sum x)^2}{n} = \sum xy - \frac{\sum x \sum y}{n}$$

$$b_1 \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] = \sum xy - \frac{\sum x \sum y}{n}$$

$$b_1 = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}} = \frac{SS_{xy}}{SS_{xx}}$$